

Lab 4, Act 2

The model is wrong. Does it matter?

NRES 746 – name: _____ group: _____

Picking up where we left off

In Act 1 the analyst fit a straight line to the *log* counts, which matches how the data were generated, and your simulations showed the slope estimate was unbiased with the spread you predicted. Real analysts do not get to pick the analysis that matches the truth. Suppose a second analyst, following the agency's old spreadsheet protocol, fits a straight line to the *raw* counts instead. That model says the index falls by the same *number* of animals every year. The truth says it falls by the same *proportion*.

We will compare the two analysts under two truths. In both, the expected index in year 0 is $N_0 = 60$, the scatter is $\sigma = 0.2$ on the log scale, and surveys are annual ($t = 1, \dots, 25$):

Scenario	Total decline, year 0 to year 25
Mild	25% over 25 years
Severe	90% over 25 years

Both analysts report the same management quantity: the estimated **percent decline over the 25 years**. The log-scale analyst computes it as $1 - e^{25\hat{b}}$. The raw-scale analyst computes it as $1 - \hat{y}_{25}/\hat{y}_0$, using the fitted values of the straight line at $t = 25$ and $t = 0$.

Does the wrong model give the wrong answer? And can the data alone tell us the model is wrong?

Before you do anything else: close your laptop and put your phone away. Parts 1-3 and the group check-in that follows are pen-and-paper only.

Part 1: What does each model assume about scatter? (individual, ~10 min)

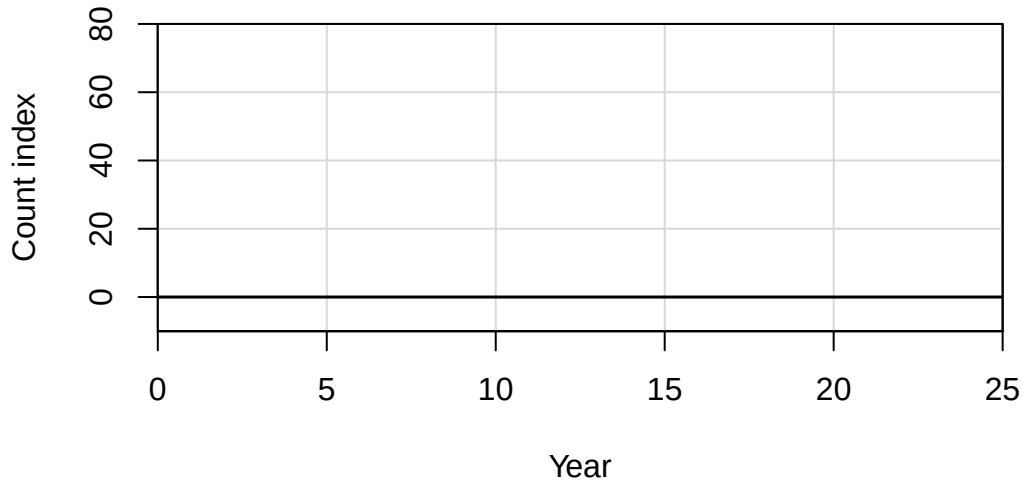
1a. The *variance function* of a distribution says how the variance changes with the mean μ . Fill in the blanks. A gamma distribution with shape α and scale s has mean αs and variance αs^2 . Bolker Ch. 4 is fair game.

Distribution	Var(Y) in terms of the mean	Constant CV?
Normal (SD sigma)		
Poisson		
Negative binomial (mean, size k)		
Gamma (mean, shape alpha)		
Lognormal (log-SD s)	$(\exp(s^2) - 1) \times \text{mean}^2$	

1b. For a lognormal variable whose log has mean m and SD s , the mean of the variable is $e^{m+s^2/2}$ and its SD is $\sqrt{e^{s^2} - 1}$ times that mean. In our model $s = \sigma = 0.2$ and the median in year 0 is $e^m = 60$. Fill in the table (a calculator is fine).

Year and scenario	Median	Mean	SD
Year 0	60		
Year 25, mild	45		
Year 25, severe	6		

1c. For the severe scenario, use the axes below to sketch (i) the median trajectory, (ii) a band showing roughly ± 2 SD around it, and (iii) the straight line an analyst would fit through the counts. Where does your line cross zero, and why is that a problem for a count?



Part 2: Commit to predictions (individual, ~4 min)

2a. For each analyst and scenario, will the estimated percent decline be unbiased, too high, or too low? Write a word or two in each cell. Then say briefly why.

	Mild (25%)	Severe (90%)
Log-scale analyst		
Raw-scale analyst		

Part 3: A check that does not need the truth (individual, ~5 min)

In real life we never know the true model. What we can do is ask whether the data look like they could have come from the *fitted* model: fit the model, simulate many new datasets from the fitted model, and compare a summary statistic of the real data with the same statistic across the simulated datasets.

Our statistic is the number of **sign changes** in the residuals over the 25 years (the number of times neighboring residuals switch between positive and negative). If the fitted line captures the pattern and the residuals are independent, about half of the 24 neighboring pairs should switch. If the true pattern is curved and the line is straight, residuals come in long runs of one sign, and very few switch.

3a. Write, in plain English, the procedure for a check that returns a *p*-value: the fraction of simulated datasets with as few sign changes as the real data. Do not write R code. Be specific about what you fit to each simulated dataset and why.

3b. How often do you expect this check to flag each model in each scenario (rarely, sometimes, or usually)?

	Mild (25%)	Severe (90%)
Log-scale model		
Raw-scale model		

Group check-in: finalize your plan, then choose your path (group, ~8 min)

Compare your answers to Parts 1-3 with your group. Agree on one procedure for the check in 3a. Where your predictions in 2a and 3b differ, argue it out.

Your group must now pick **one** of the two paths below for running the experiments. Both are equally valid; decide as a group before opening any laptops.

- **Path A (AI-assisted):** write a prompt describing your procedure and the model, and have an AI assistant generate the R code.
- **Path B (self-coded):** write the R code yourselves, using R's documentation and each other, but no AI tools.

Our group is using: Path A (AI-assisted) / Path B (self-coded) (*circle one*)

Laptops open now. Both paths run the same three experiments. Choose the instructions for your path.

Part 4: Run the experiments (laptops, ~30 min)

Use these settings: $N_0 = 60$, $\sigma = 0.2$, $t = 1, \dots, 25$, $b_{\text{mild}} = \ln(0.75)/25$ and $b_{\text{severe}} = \ln(0.10)/25$. The two helper functions below are ready-made. Type them into your R script.

```
# number of times neighboring residuals switch sign
sign_changes <- function(r) {
  s <- sign(r)
  sum(s[-1] != s[-length(s)])
}

# t: years, y: observed counts, sims: matrix of simulated counts
# (one row per year, one column per simulated data set)
plot_envelope <- function(t, y, sims) {
  lo <- apply(sims, 1, quantile, 0.05)
  hi <- apply(sims, 1, quantile, 0.95)
  plot(t, y, type = "n", ylim = range(y, lo, hi), xlab = "Year", ylab = "Count index")
  polygon(c(t, rev(t)), c(lo, rev(hi)), col = "grey85", border = NA)
  points(t, y, pch = 16)
  abline(h = 0, lty = 3)
}
```

The three experiments:

1. **Two analysts.** For each scenario, simulate 2,000 datasets from the model. Fit both analysts to every dataset and compute each one's estimated percent decline. Record the mean and SD of the estimates and the share of estimates above 100%.
2. **An envelope.** Simulate one dataset from the severe scenario and fit the raw-scale model. Simulate 500 new datasets from the fitted raw-scale model (Normal with mean equal to the fitted values and SD equal to the residual SD). Draw the envelope with `plot_envelope()`.
3. **The check.** Implement your procedure from 3a as a function. For the log-scale model, run the same check on $\log(y)$, with the simulated datasets also generated on the log scale. Use 200 simulated datasets inside the check. Apply it to 100 fresh datasets for each scenario and each model, and record how often it returns $p < 0.05$.

Path A: AI-assisted

Write a prompt describing your group's procedure and the model, and asking the AI to write R code for the three experiments. Record your prompt and what you had to change to make the code do what you intended (do not paste the AI's code).

Reflection. Did the AI's first attempt do what your procedure said? Where did your own understanding of the models from Parts 1-3 help you catch or fix something?

Path B: Self-coded

Write the R code yourselves (R's built-in help, documentation, and human-written websites and neighbors are fair game; AI tools are not).

Reflection. What part of turning your procedure into working R gave your group the most trouble? How did you resolve it?

Results (both paths)

Experiment 1. Estimated percent decline over 25 years.

Scenario	Analyst	Mean	SD	Share > 100%
Mild (25%)	Log-scale			
Mild (25%)	Raw-scale			
Severe (90%)	Log-scale			
Severe (90%)	Raw-scale			

Experiment 3. Share of datasets for which the check returns $p < 0.05$.

	Mild (25%)	Severe (90%)
Log-scale model		
Raw-scale model		

In one or two sentences, describe what the envelope in Experiment 2 shows that a single glance at the raw counts might not.

If you finish early: simulate 100,000 values of y_0 from the model. Compare $\text{mean}(\log(y_0))$ with $\log(\text{mean}(y_0))$. The gap should match $\sigma^2/2$ from 1b. Why does the average of the logs differ from the log of the average?

Group discussion and revision (~5 min)

Compare your results with a different group (ideally one that took the other path). How did your predictions in 2a and 3b compare with the results? In which scenario did the wrong model matter, and could the check have warned you?

Feel free to revise any of your answers above based on the discussion (cross out and rewrite; I like to see your thinking evolve).

Please take a minute to record any changes you made and any new insights gained during the group discussion. You can also use this space to reflect on any areas of confusion (“muddiest points”) or topics you want to review or dust off.
